

Summary report of CHAEA learning styles by CHAEA³S package

This report contains the most important results of the analysis that is conducted to unveil the learning styles that are present in the group of students under study. The analysis is based on the learning styles considered by CHAEA: activist, reflector, theorist, and pragmatist. Unless otherwise stated, the uncertainties throughout the document (in parenthesis) have been obtained using a t-Student distribution with a confidence interval of 95%.

The report is structured as follows. First, Section 1 is devoted to the individual and global statistical analysis. Here, the importance of the different learning styles for each individual student can be found (both quantitatively as well as qualitatively). Then, average means and confidence intervals are presented, along with the affinities and the Probability Density Functions. Second, Section 2 discusses the principal component analysis. The eigenvalues and eigenvectors of the covariance matrix are first introduced. Subsequently, the learning styles of the students in the principal components basis set is presented. Next, a reduced dimensional representation of the data is conducted. Third, the participation ratios are presented in Section 3. The values for the original (activist, reflector, theorist, and pragmatist) and in the principal components basis sets are listed. Here, a statistical analysis of the distribution of the participation ratios is performed. The report concludes in Section 4 with a brief summary of the main characteristics of the students, along with some recommendations.

Further information at: J. Ablanque, V. Gabaldon, P. Almendros, J. C. Losada, R. M. Benito, and F. Revuelta. CHAEA3S: A software package for comprehensive analysis of learning styles and academic performance. PLOS ONE (2026).

1. Individual and global statistical analysis

In this section we present the individual and global statistical analysis of the learning styles as originally defined in CHAEA (activist, reflector, theorist, and pragmatist). This section is divided in two parts. First, the (quantitative and qualitative) importance of the different learning styles for each individual student can be found. Second, a global analysis is performed, where average means, confidence intervals, affinities, and the probability density functions can be found.

1.1 Individual analysis

Quantitative description of the learning styles for each individual student

Table 1 shows the number of points (from 0 to 20) that each student gets in CHAEA for the different learning styles (only the first 150 students are included).

Table 1. Points related the learning styles for each of the students.

Student	Activist	Reflector	Theorist	Pragmatist
fisicaI_2022-23_01.xls	14.0	14.0	14.0	15.0
fisicaI_2022-23_02.xls	13.0	18.0	12.0	10.0
fisicaI_2022-23_03.xls	13.0	15.0	14.0	13.0
fisicaI_2022-23_04.xlsx	9.0	17.0	13.0	8.0
fisicaI_2022-23_05.xlsx	15.0	18.0	12.0	13.0
fisicaI_2022-23_06.xls	17.0	13.0	16.0	12.0
fisicaI_2022-23_07.xlsx	14.0	17.0	11.0	11.0
fisicaI_2022-23_08.xls	8.0	18.0	20.0	15.0
fisicaI_2022-23_09.xls	6.0	15.0	15.0	12.0
fisicaI_2022-23_11.xls	14.0	16.0	12.0	14.0
fisicaI_2022-	17.0	15.0	19.0	18.0

23_12.xls				
fisicaI_2022-23_13.xlsx	17.0	15.0	13.0	17.0
fisicaI_2022-23_14.xls	6.0	17.0	8.0	9.0
fisicaI_2022-23_15.xlsx	10.0	13.0	11.0	10.0
fisicaI_2022-23_17.xls	9.0	16.0	12.0	13.0
fisicaI_2022-23_18.xls	9.0	19.0	14.0	13.0
fisicaI_2022-23_19.xls	10.0	16.0	15.0	16.0
fisicaI_2022-23_20.xls	10.0	12.0	13.0	13.0
fisicaI_2022-23_21.xls	15.0	16.0	11.0	15.0
fisicaI_2022-23_22.xlsx	15.0	13.0	11.0	19.0
fisicaI_2022-23_23.xls	11.0	18.0	17.0	12.0
fisicaI_2022-23_24.xls	10.0	12.0	13.0	16.0
fisicaI_2022-23_25.xls	12.0	15.0	13.0	10.0
fisicaI_2022-23_26.xls	13.0	16.0	11.0	11.0
fisicaI_2022-23_27.xls	13.0	19.0	15.0	14.0
fisicaI_2022-23_30.xls	9.0	16.0	12.0	14.0

fisicaI_2022-23_31.xls	15.0	19.0	9.0	15.0
fisicaI_2022-23_33.xlsx	10.0	9.0	11.0	14.0
fisicaI_2022-23_34.xls	15.0	10.0	13.0	20.0
fisicaI_2022-23_35.xls	16.0	18.0	8.0	9.0
fisicaI_2022-23_36.xls	10.0	15.0	11.0	14.0
fisicaI_2022-23_37.xls	8.0	15.0	12.0	13.0
fisicaI_2022-23_38.xlsx	10.0	15.0	8.0	7.0
fisicaI_2022-23_39.xls	4.0	20.0	18.0	13.0
fisicaI_2022-23_40.xls	9.0	19.0	11.0	19.0
fisicaI_2022-23_41.xls	11.0	15.0	16.0	16.0
fisicaI_2022-23_42.xls	11.0	17.0	15.0	12.0
fisicaI_2022-23_43.xls	9.0	16.0	14.0	13.0
fisicaI_2022-23_44.xls	12.0	15.0	13.0	14.0
fisicaI_2022-23_45.xlsx	14.0	13.0	15.0	12.0
fisicaI_2022-23_46.xls	8.0	18.0	12.0	12.0
fisicaI_2022-23_47.xls	11.0	15.0	10.0	10.0
fisicaI_2022-	8.0	19.0	19.0	15.0

23_48.xls

fisicaI_2022-23_49.xls	12.0	16.0	13.0	12.0
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fisicaI_2022-23_50.xls	5.0	19.0	17.0	11.0
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fisicaI_2022-23_51.xls	2.0	16.0	14.0	7.0
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fisicaI_2022-23_52.xls	6.0	16.0	13.0	11.0
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fisicaI_2022-23_53.xlsx	12.0	18.0	14.0	17.0
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fisicaI_2022-23_54.xlsx	11.0	17.0	15.0	12.0
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fisicaI_2022-23_55.xls	12.0	16.0	15.0	10.0
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fisicaI_2022-23_56.xls	11.0	18.0	10.0	12.0
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fisicaI_2022-23_57.xls	13.0	12.0	13.0	14.0
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fisicaI_2022-23_58.xlsx	12.0	17.0	16.0	12.0
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fisicaI_2022-23_59.xls	5.0	15.0	13.0	8.0
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fisicaI_2022-23_61.xls	12.0	12.0	10.0	12.0
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fisicaI_2022-23_62.xls	7.0	14.0	14.0	12.0
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fisicaI_2022-23_63.xlsx	10.0	16.0	14.0	11.0
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fisicaI_2022-23_64.xlsx	7.0	13.0	8.0	8.0
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fisicaI_2022-23_65.xls	9.0	13.0	16.0	14.0
fisicaI_2022-23_66.xls	6.0	15.0	9.0	7.0
fisicaI_2022-23_67.xls	12.0	14.0	11.0	15.0
fisicaI_2022-23_68.xls	13.0	15.0	11.0	9.0
fisicaI_2022-23_69.xls	15.0	14.0	12.0	12.0
fisicaI_2022-23_70.xls	13.0	13.0	6.0	7.0
fisicaI_2022-23_71.xls	0.0	0.0	0.0	0.0
fisicaI_2022-23_72.xls	13.0	17.0	16.0	12.0
fisicaI_2022-23_73.xls	12.0	17.0	14.0	14.0
fisicaI_2022-23_74.xlsx	6.0	15.0	15.0	7.0
fisicaI_2022-23_75.xls	12.0	12.0	14.0	10.0
fisicaI_2022-23_76.xls	14.0	10.0	7.0	10.0
fisicaI_2023-24_01.xls	13.0	16.0	10.0	13.0
fisicaI_2023-24_02.xls	10.0	17.0	18.0	19.0
fisicaI_2023-24_03.xlsx	14.0	17.0	18.0	15.0
fisicaI_2023-24_04.xls	17.0	16.0	12.0	13.0
fisicaI_2023-	17.0	11.0	9.0	9.0

24_05.xls

fisicaI_2023-24_06.xlsx	9.0	20.0	17.0	14.0
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fisicaI_2023-24_07.xlsx	8.0	18.0	17.0	14.0
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fisicaI_2023-24_08.xls	15.0	11.0	9.0	14.0
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fisicaI_2023-24_09.xls	13.0	15.0	14.0	16.0
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fisicaI_2023-24_10.xls	9.0	17.0	13.0	13.0
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fisicaI_2023-24_11.xlsx	7.0	16.0	15.0	11.0
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fisicaI_2023-24_12.xls	9.0	13.0	16.0	14.0
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fisicaI_2023-24_13.xls	10.0	18.0	12.0	8.0
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fisicaI_2023-24_14.xls	11.0	15.0	13.0	12.0
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fisicaI_2023-24_15.xls	11.0	20.0	15.0	10.0
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fisicaI_2023-24_16.xls	8.0	19.0	15.0	13.0
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fisicaI_2023-24_17.xls	12.0	17.0	15.0	10.0
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fisicaI_2023-24_18.xls	9.0	16.0	16.0	18.0
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fisicaI_2023-24_20.xls	3.0	17.0	18.0	12.0
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fisicaI_2023-24_21.xls	7.0	17.0	15.0	11.0
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fisicaI_2023-24_22.xls	10.0	9.0	15.0	9.0
fisicaI_2023-24_23.xls	9.0	15.0	14.0	11.0
fisicaI_2023-24_24.xls	14.0	13.0	8.0	15.0
fisicaI_2023-24_25.xls	16.0	11.0	15.0	15.0
fisicaI_2023-24_26.xls	8.0	18.0	17.0	10.0
fisicaI_2023-24_27.xls	15.0	18.0	18.0	18.0
fisicaI_2023-24_28.xls	6.0	20.0	17.0	13.0
fisicaI_2023-24_29.xlsx	6.0	1.0	3.0	6.0
fisicaI_2023-24_30.xls	14.0	17.0	15.0	13.0
fisicaI_2023-24_31.xls	17.0	17.0	15.0	20.0
fisicaI_2023-24_32.xls	12.0	11.0	18.0	11.0
fisicaI_2023-24_33.xls	15.0	17.0	13.0	13.0
fisicaI_2023-24_34.xls	16.0	13.0	14.0	18.0
fisicaI_2023-24_35.xls	11.0	19.0	12.0	12.0
fisicaI_2023-24_36.xls	12.0	16.0	15.0	16.0
fisicaI_2023-24_37.xls	14.0	13.0	7.0	11.0
fisicaI_2023-	6.0	17.0	16.0	15.0

24_38.xlsx

fisicaI_2023-24_39.xlsx	15.0	19.0	14.0	14.0
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fisicaI_2023-24_40.xls	10.0	18.0	14.0	13.0
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fisicaI_2023-24_41.xls	2.0	17.0	13.0	12.0
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fisicaI_2023-24_42.xls	15.0	12.0	10.0	13.0
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fisicaI_2023-24_43.xls	9.0	19.0	15.0	11.0
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fisicaI_2023-24_44.xlsx	3.0	17.0	15.0	9.0
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fisicaI_2023-24_45.xls	8.0	18.0	16.0	17.0
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fisicaI_2023-24_46.xls	5.0	20.0	14.0	11.0
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fisicaI_2023-24_47.xls	10.0	13.0	15.0	14.0
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fisicaI_2023-24_48.xls	16.0	13.0	11.0	16.0
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fisicaI_2023-24_49.xlsx	9.0	15.0	12.0	14.0
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fisicaI_2023-24_50.xlsx	5.0	17.0	16.0	12.0
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fisicaI_2023-24_51.xls	13.0	17.0	10.0	12.0
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fisicaI_2023-24_52.xlsx	11.0	15.0	15.0	14.0
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fisicaI_2023-24_53.xlsx	11.0	18.0	16.0	14.0
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fisicaI_2023-24_54.xlsx	11.0	15.0	14.0	13.0
fisicaI_2023-24_55.xls	13.0	12.0	16.0	17.0
fisicaI_2023-24_56.xls	12.0	13.0	15.0	10.0
fisicaI_2023-24_57.xlsx	11.0	15.0	16.0	13.0
fisicaI_2023-24_58.xls	9.0	19.0	18.0	16.0
fisicaI_2023-24_59.xls	14.0	18.0	16.0	16.0
fisicaI_2023-24_60.xlsx	14.0	18.0	18.0	18.0
fisicaI_2024-25_01.xls	9.0	19.0	18.0	17.0
fisicaI_2024-25_02.xls	11.0	19.0	19.0	18.0
fisicaI_2024-25_03.xlsx	15.0	17.0	16.0	15.0
fisicaI_2024-25_04.xls	11.0	15.0	13.0	13.0
fisicaI_2024-25_05.xls	13.0	15.0	8.0	11.0
fisicaI_2024-25_06.xls	16.0	15.0	12.0	13.0
fisicaI_2024-25_07.xls	6.0	16.0	14.0	11.0
fisicaI_2024-25_08.xls	15.0	16.0	12.0	15.0
fisicaI_2024-25_09.xls	14.0	15.0	15.0	14.0
fisicaI_2024-	13.0	11.0	15.0	10.0

25_10.xls				
fisicaI_2024-25_100.xls	10.0	18.0	14.0	14.0
fisicaI_2024-25_101.xls	11.0	17.0	12.0	11.0
fisicaI_2024-25_102.xlsx	15.0	17.0	16.0	18.0
fisicaI_2024-25_103.xls	10.0	14.0	9.0	12.0
fisicaI_2024-25_104.xls	9.0	17.0	12.0	12.0
fisicaI_2024-25_105.xls	11.0	19.0	16.0	12.0
fisicaI_2024-25_11.xlsx	15.0	13.0	13.0	11.0
fisicaI_2024-25_12.xlsx	4.0	16.0	13.0	7.0
fisicaI_2024-25_13.xls	12.0	17.0	12.0	11.0
fisicaI_2024-25_14.xlsx	9.0	18.0	13.0	8.0
fisicaI_2024-25_15.xls	14.0	16.0	9.0	11.0

The following students were not considered in the analysis because of errors in the CHAEA files:

fisicaI_2024-25_89.xls.

Qualitative description of the learning styles for each individual student

Table 2 shows the individual qualitative tendency (very low, low, moderate, high, or very high) that each student has towards each of the learning styles (only the first 150 students are included).

Table 2. Tendency towards the learning styles for each of the students.

Student	Activist	Reflector	Theorist	Pragmatist
fisicaI_2022-23_01.xls	High	Moderate	High	High
fisicaI_2022-23_02.xls	High	High	Moderate	Low
fisicaI_2022-23_03.xls	High	Moderate	High	Moderate
fisicaI_2022-23_04.xlsx	Moderate	Moderate	Moderate	Very low
fisicaI_2022-23_05.xlsx	Very high	High	Moderate	Moderate
fisicaI_2022-23_06.xls	Very high	Low	Very high	Moderate
fisicaI_2022-23_07.xlsx	High	Moderate	Moderate	Moderate
fisicaI_2022-23_08.xls	Low	High	Very high	High
fisicaI_2022-23_09.xls	Very low	Moderate	High	Moderate
fisicaI_2022-23_11.xls	High	Moderate	Moderate	High
fisicaI_2022-23_12.xls	Very high	Moderate	Very high	Very high
fisicaI_2022-23_13.xlsx	Very high	Moderate	Moderate	Very high
fisicaI_2022-23_14.xls	Very low	Moderate	Low	Low
fisicaI_2022-	Moderate	Low	Moderate	Low

23_15.xlsx

fisicaI_2022-23_17.xls	Moderate	Moderate	Moderate	Moderate
fisicaI_2022-23_18.xls	Moderate	High	High	Moderate
fisicaI_2022-23_19.xls	Moderate	Moderate	High	Very high
fisicaI_2022-23_20.xls	Moderate	Low	Moderate	Moderate
fisicaI_2022-23_21.xls	Very high	Moderate	Moderate	High
fisicaI_2022-23_22.xlsx	Very high	Low	Moderate	Very high
fisicaI_2022-23_23.xls	Moderate	High	Very high	Moderate
fisicaI_2022-23_24.xls	Moderate	Low	Moderate	Very high
fisicaI_2022-23_25.xls	Moderate	Moderate	Moderate	Low
fisicaI_2022-23_26.xls	High	Moderate	Moderate	Moderate
fisicaI_2022-23_27.xls	High	High	High	High
fisicaI_2022-23_30.xls	Moderate	Moderate	Moderate	High
fisicaI_2022-23_31.xls	Very high	High	Low	High
fisicaI_2022-23_33.xlsx	Moderate	Very low	Moderate	High
fisicaI_2022-23_34.xls	Very high	Very low	Moderate	Very high

fisicaI_2022-23_35.xls	Very high	High	Low	Low
fisicaI_2022-23_36.xls	Moderate	Moderate	Moderate	High
fisicaI_2022-23_37.xls	Low	Moderate	Moderate	Moderate
fisicaI_2022-23_38.xlsx	Moderate	Moderate	Low	Very low
fisicaI_2022-23_39.xls	Very low	Very high	Very high	Moderate
fisicaI_2022-23_40.xls	Moderate	High	Moderate	Very high
fisicaI_2022-23_41.xls	Moderate	Moderate	Very high	Very high
fisicaI_2022-23_42.xls	Moderate	Moderate	High	Moderate
fisicaI_2022-23_43.xls	Moderate	Moderate	High	Moderate
fisicaI_2022-23_44.xls	Moderate	Moderate	Moderate	High
fisicaI_2022-23_45.xlsx	High	Low	High	Moderate
fisicaI_2022-23_46.xls	Low	High	Moderate	Moderate
fisicaI_2022-23_47.xls	Moderate	Moderate	Moderate	Low
fisicaI_2022-23_48.xls	Low	High	Very high	High
fisicaI_2022-23_49.xls	Moderate	Moderate	Moderate	Moderate
fisicaI_2022-23_50.xls	Very low	High	Very high	Moderate
fisicaI_2022-	Very low	Moderate	High	Very low

23_51.xls				
fisicaI_2022-23_52.xls	Very low	Moderate	Moderate	Moderate
fisicaI_2022-23_53.xlsx	Moderate	High	High	Very high
fisicaI_2022-23_54.xlsx	Moderate	Moderate	High	Moderate
fisicaI_2022-23_55.xls	Moderate	Moderate	High	Low
fisicaI_2022-23_56.xls	Moderate	High	Moderate	Moderate
fisicaI_2022-23_57.xls	High	Low	Moderate	High
fisicaI_2022-23_58.xlsx	Moderate	Moderate	Very high	Moderate
fisicaI_2022-23_59.xls	Very low	Moderate	Moderate	Very low
fisicaI_2022-23_61.xls	Moderate	Low	Moderate	Moderate
fisicaI_2022-23_62.xls	Low	Moderate	High	Moderate
fisicaI_2022-23_63.xlsx	Moderate	Moderate	High	Moderate
fisicaI_2022-23_64.xlsx	Low	Low	Low	Very low
fisicaI_2022-23_65.xls	Moderate	Low	Very high	High
fisicaI_2022-23_66.xls	Very low	Moderate	Low	Very low
fisicaI_2022-23_67.xls	Moderate	Moderate	Moderate	High

fisicaI_2022-23_68.xls	High	Moderate	Moderate	Low
fisicaI_2022-23_69.xls	Very high	Moderate	Moderate	Moderate
fisicaI_2022-23_70.xls	High	Low	Very low	Very low
fisicaI_2022-23_71.xls	Very low	Very low	Very low	Very low
fisicaI_2022-23_72.xls	High	Moderate	Very high	Moderate
fisicaI_2022-23_73.xls	Moderate	Moderate	High	High
fisicaI_2022-23_74.xlsx	Very low	Moderate	High	Very low
fisicaI_2022-23_75.xls	Moderate	Low	High	Low
fisicaI_2022-23_76.xls	High	Very low	Low	Low
fisicaI_2023-24_01.xls	High	Moderate	Moderate	Moderate
fisicaI_2023-24_02.xls	Moderate	Moderate	Very high	Very high
fisicaI_2023-24_03.xlsx	High	Moderate	Very high	High
fisicaI_2023-24_04.xls	Very high	Moderate	Moderate	Moderate
fisicaI_2023-24_05.xls	Very high	Low	Low	Low
fisicaI_2023-24_06.xlsx	Moderate	Very high	Very high	High
fisicaI_2023-24_07.xlsx	Low	High	Very high	High
fisicaI_2023-	Very high	Low	Low	High

24_08.xls

fisicaI_2023-24_09.xls	High	Moderate	High	Very high
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fisicaI_2023-24_10.xls	Moderate	Moderate	Moderate	Moderate
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fisicaI_2023-24_11.xlsx	Low	Moderate	High	Moderate
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fisicaI_2023-24_12.xls	Moderate	Low	Very high	High
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fisicaI_2023-24_13.xls	Moderate	High	Moderate	Very low
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fisicaI_2023-24_14.xls	Moderate	Moderate	Moderate	Moderate
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fisicaI_2023-24_15.xls	Moderate	Very high	High	Low
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fisicaI_2023-24_16.xls	Low	High	High	Moderate
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fisicaI_2023-24_17.xls	Moderate	Moderate	High	Low
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fisicaI_2023-24_18.xls	Moderate	Moderate	Very high	Very high
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fisicaI_2023-24_20.xls	Very low	Moderate	Very high	Moderate
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fisicaI_2023-24_21.xls	Low	Moderate	High	Moderate
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fisicaI_2023-24_22.xls	Moderate	Very low	High	Low
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fisicaI_2023-24_23.xls	Moderate	Moderate	High	Moderate
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fisicaI_2023-24_24.xls	High	Low	Low	High
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fisicaI_2023-24_25.xls	Very high	Low	High	High
fisicaI_2023-24_26.xls	Low	High	Very high	Low
fisicaI_2023-24_27.xls	Very high	High	Very high	Very high
fisicaI_2023-24_28.xls	Very low	Very high	Very high	Moderate
fisicaI_2023-24_29.xlsx	Very low	Very low	Very low	Very low
fisicaI_2023-24_30.xls	High	Moderate	High	Moderate
fisicaI_2023-24_31.xls	Very high	Moderate	High	Very high
fisicaI_2023-24_32.xls	Moderate	Low	Very high	Moderate
fisicaI_2023-24_33.xls	Very high	Moderate	Moderate	Moderate
fisicaI_2023-24_34.xls	Very high	Low	High	Very high
fisicaI_2023-24_35.xls	Moderate	High	Moderate	Moderate
fisicaI_2023-24_36.xls	Moderate	Moderate	High	Very high
fisicaI_2023-24_37.xls	High	Low	Low	Moderate
fisicaI_2023-24_38.xlsx	Very low	Moderate	Very high	High
fisicaI_2023-24_39.xlsx	Very high	High	High	High
fisicaI_2023-24_40.xls	Moderate	High	High	Moderate
fisicaI_2023-	Very low	Moderate	Moderate	Moderate

24_41.xls

fisicaI_2023-24_42.xls	Very high	Low	Moderate	Moderate
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fisicaI_2023-24_43.xls	Moderate	High	High	Moderate
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fisicaI_2023-24_44.xlsx	Very low	Moderate	High	Low
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fisicaI_2023-24_45.xls	Low	High	Very high	Very high
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fisicaI_2023-24_46.xls	Very low	Very high	High	Moderate
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fisicaI_2023-24_47.xls	Moderate	Low	High	High
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fisicaI_2023-24_48.xls	Very high	Low	Moderate	Very high
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fisicaI_2023-24_49.xlsx	Moderate	Moderate	Moderate	High
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fisicaI_2023-24_50.xlsx	Very low	Moderate	Very high	Moderate
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fisicaI_2023-24_51.xls	High	Moderate	Moderate	Moderate
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fisicaI_2023-24_52.xlsx	Moderate	Moderate	High	High
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fisicaI_2023-24_53.xlsx	Moderate	High	Very high	High
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fisicaI_2023-24_54.xlsx	Moderate	Moderate	High	Moderate
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fisicaI_2023-24_55.xls	High	Low	Very high	Very high
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fisicaI_2023-24_56.xls	Moderate	Low	High	Low
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fisicaI_2023-24_57.xlsx	Moderate	Moderate	Very high	Moderate
fisicaI_2023-24_58.xls	Moderate	High	Very high	Very high
fisicaI_2023-24_59.xls	High	High	Very high	Very high
fisicaI_2023-24_60.xlsx	High	High	Very high	Very high
fisicaI_2024-25_01.xls	Moderate	High	Very high	Very high
fisicaI_2024-25_02.xls	Moderate	High	Very high	Very high
fisicaI_2024-25_03.xlsx	Very high	Moderate	Very high	High
fisicaI_2024-25_04.xls	Moderate	Moderate	Moderate	Moderate
fisicaI_2024-25_05.xls	High	Moderate	Low	Moderate
fisicaI_2024-25_06.xls	Very high	Moderate	Moderate	Moderate
fisicaI_2024-25_07.xls	Very low	Moderate	High	Moderate
fisicaI_2024-25_08.xls	Very high	Moderate	Moderate	High
fisicaI_2024-25_09.xls	High	Moderate	High	High
fisicaI_2024-25_10.xls	High	Low	High	Low
fisicaI_2024-25_100.xls	Moderate	High	High	High
fisicaI_2024-25_101.xls	Moderate	Moderate	Moderate	Moderate
fisicaI_2024-	Very high	Moderate	Very high	Very high

25_102.xlsx				
fisicaI_2024-25_103.xls	Moderate	Moderate	Low	Moderate
fisicaI_2024-25_104.xls	Moderate	Moderate	Moderate	Moderate
fisicaI_2024-25_105.xls	Moderate	High	Very high	Moderate
fisicaI_2024-25_11.xlsx	Very high	Low	Moderate	Moderate
fisicaI_2024-25_12.xlsx	Very low	Moderate	Moderate	Very low
fisicaI_2024-25_13.xls	Moderate	Moderate	Moderate	Moderate
fisicaI_2024-25_14.xlsx	Moderate	High	Moderate	Very low
fisicaI_2024-25_15.xls	High	Moderate	Low	Moderate

1.2 Global analysis

Global tendencies towards the learning styles of the students (I)

Table 3 presents the number and percentage of students with the same tendency (very low, low, moderate, high, or very high) towards each of the learning styles. The bottom line shows the average tendency. These results are also shown as a barr graphic in Fig. 1.

Table 3. Number and percentage of students with the same tendency towards each of the learning styles.

Tendency	Very low		Low		Moderate		High		Very high	
	No.	%	No.	%	No.	%	No.	%	No.	%
Activist	31	13.4	18	7.8	95	41.1	46	19.9	41	17.7
Reflector	13	5.6	43	18.6	121	52.4	47	20.3	7	3.0
Theorist	8	3.5	20	8.7	84	36.4	58	25.1	61	26.4
Pragmatist	19	8.2	27	11.7	94	40.7	48	20.8	43	18.6
Total	71	9.5	108	13.2	394	43.5	199	21.7	152	20.8

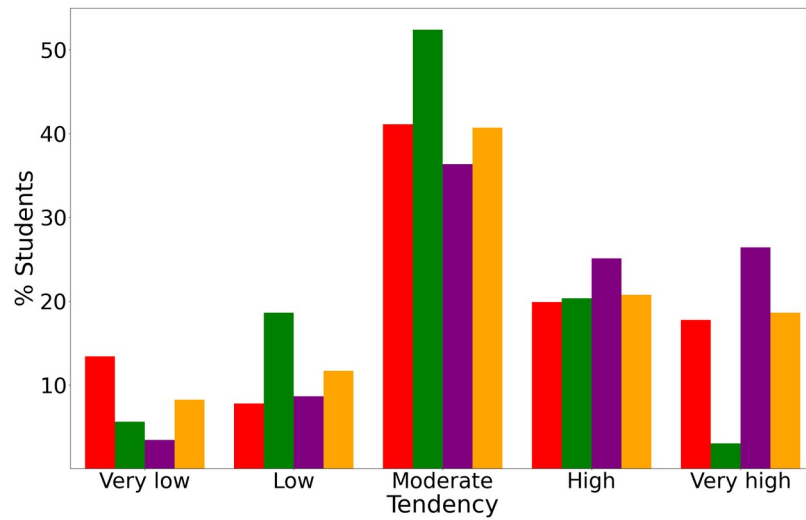


Figure 1. Bar graphic showing the percentage of students with a very low (leftmost), low (left), moderate (middle), high (right) and very high (rightmost) tendency towards the activist (red), reflector (green), theorist (purple) and pragmatist (orange) learning styles. The shown results correspond to the values of Table 3.

Global tendencies towards the learning styles of the students (II)

Table 4 shows the average mean and the uncertainties for the different learning styles, along with the corresponding qualitative tendency, and the corresponding affinity. For the shake of clarity, the average profile of the students given by the average means and the corresponding uncertainties is shown in Fig. 2, while the affinities are represented as a barr graphic in Fig. 3.

Table 4. Numerical average values of each learning style with the uncertainties (in parenthesis) along with their qualitative tendency, and the corresponding affinity.

Learning style	Average mean (Uncertainty)	Tendency	Affinity (%)
Activist	11.0(0.4)	Moderate	78.8
Reflector	15.2(0.4)	Moderate	75.8
Theorist	13.2(0.4)	Moderate/ High	87.9
Pragmatist	12.7(0.4)	Moderate	80.1

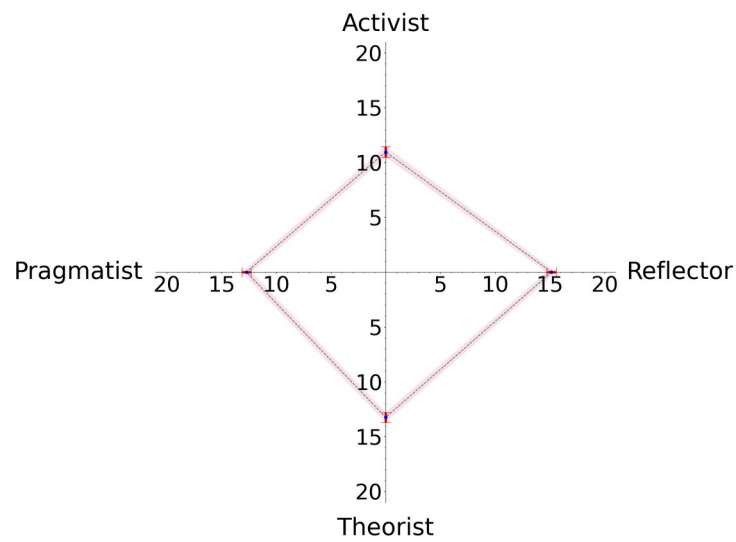


Figure 2. Average profile of the learning styles of the students.

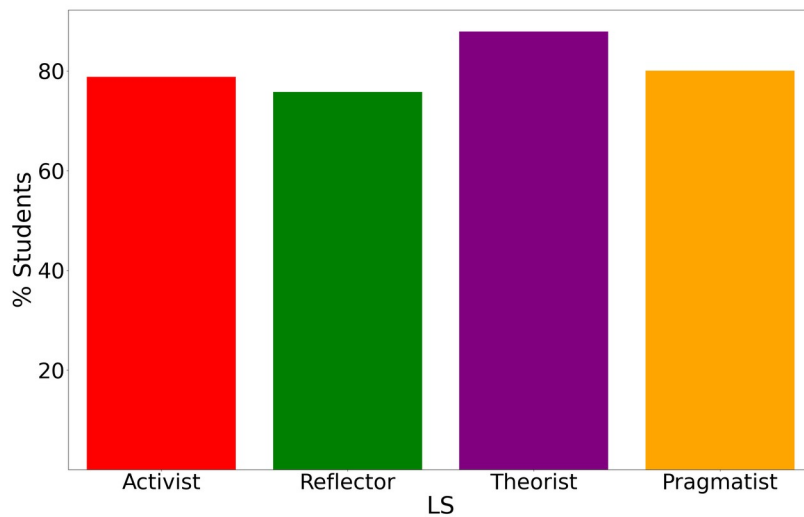


Figure 3. Affinity of the learning styles. The bar graphic gives the percentage of students that have a noticeable tendency towards the activist (red), reflector (green), theorist (purple) and pragmatist (orange) learning styles.

Figure 4 shows histograms with the probability distributions of the results of Table 1. The continuous lines show fittings with Weibull distributions with the parameters shown in Table 5. These fittings have been performed on the corresponding cumulative distributions given by the staircases shown in Figure 5.

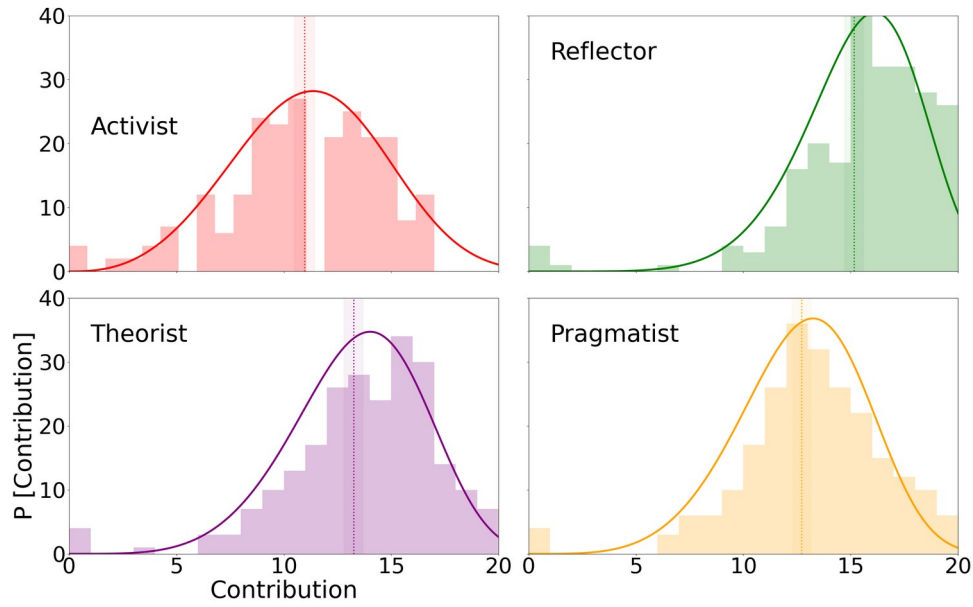


Figure 4. Histograms showing the probability distributions for the activist (red), reflector (green), theorist (purple), and pragmatist (yellow) learning styles. The continuous lines show fittings provided by the Weibull distributions with the parameters contained in Table 5. The vertical dotted lines mark the average values, and the shaded areas around them the corresponding confidence intervals.

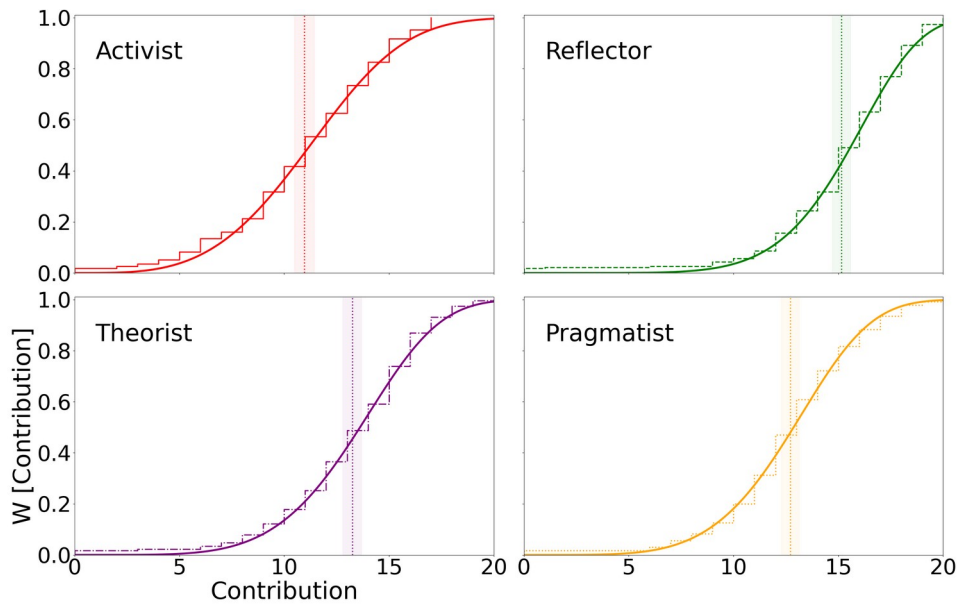


Figure 5. Same as Fig. 4 for the cumulative distributions (staircases).

Table 5. Parameters of the Weibull distributions that fit the probability distributions (histograms) shown in Figs. 4 and 5. The location parameter is set equal to $\theta = 0$.

Learning style	α	k
Activist	12.45	3.58
Reflector	16.52	6.64
Theorist	14.65	5.05
Pragmatist	13.86	4.93

2. Principal component analysis

This section presents the principal component analysis of the learning styles. It is organized as follows. First, the eigenvalues and eigenvectors of the covariance matrix are presented. Second, the description of the learning styles of the students in the principal components basis set is discussed. Finally, a reduced dimensional representation of the data is conducted.

2.1 Structure of the principal components

Eigenvalues and dispersion of the covariance matrix

In this section, we discuss the essentials of the principal components. For this purpose, Table 6 shows the four eigenvalues of the covariance matrix. Here, not only their values are listed but also their contribution to the total dispersion (given by the trace of the covariance matrix $\text{tr}(K)=48.14$, which equals the sum of all the eigenvalues) as a percentage. Likewise, the dispersion accounted by solely the principal component with the largest eigenvalue $\Sigma_0(\%)=\lambda_0*100/\text{tr}(K)$, and by combining the principal components with the two, three, and four largest eigenvalues $\Sigma_1(\%)=(\lambda_0+\lambda_1)*100/\text{tr}(K)$, $\Sigma_2(\%)=(\lambda_0+\lambda_1+\lambda_2)*100/\text{tr}(K)$, and by four $\Sigma_3(\%)=(\lambda_0+\lambda_1+\lambda_2+\lambda_3)*100/\text{tr}(K)=100\%$, respectively, are given.

Table 6. Eigenvalues of the covariance matrix given by their corresponding values λ_i and as a percentage of the total dispersion, and percentage of total dispersion Σ_i accounted by combination of the principal components with the eigenvalues λ_j , being $j \leq i$.

Principal component (i)	λ_i	$\lambda_i(\%)$	$\Sigma_i(\%)$
0	25.29	52.5	52.5
1	14.2	29.5	82.0
2	5.51	11.4	93.5
3	3.13	6.5	100.0

Eigenvectors of the covariance matrix

Table 7 shows the structure of the eigenvectors of the covariance matrix in the basis set of CHAEA learning styles. Each cell contains the percentage of the eigenvector of the principal component 0, 1, 2, and 3 that is projected on the corresponding learning style (activist, reflector, theorist, and pragmatist).

Table 7. Structure (as percentages) of the eigenvectors of the covariance matrix in the basis set of CHAEA learning styles.

Principal component	Activist	Reflector	Theorist	Pragmatist
0	12.1	24.6	31.8	31.5
1	68.0	14.5	13.6	3.9
2	9.9	53.1	10.4	26.6
3	10.0	7.8	44.2	38.1

2.2 Individual analysis of the principal components

Quantitative description of the students in the basis set of the principal components

Table 8 shows (as percentages) the structure of the learning styles in the basis set formed by the principal components. The table also includes the percentage of the learning styles of each student that is described by combining the two (sum of the percentages for the principal components 0 and 1) or three (sum of the percentages for the principal components 0, 1, and 2) principal components with the largest eigenvalues. Recall that when the four principal components are considered, 100% of the learning style of the student is reproduced.

Table 8. Structure of the learning styles of each of the students in the basis set of principal components.

Student	0	1	2	3	0+1	0+1+2
fisicaI_202 2-23_01.xls	29.34	59.42	10.4	0.84	88.75	99.16
fisicaI_202 2-23_02.xls	0.05	1.3	96.36	2.29	1.35	97.71
fisicaI_202 2-23_03.xls	30.66	47.77	0.4	21.17	78.43	78.83
fisicaI_202 2- 23_04.xlsx	22.05	33.72	35.43	8.8	55.78	91.2
fisicaI_202 2- 23_05.xlsx	19.83	29.32	49.8	1.05	49.15	98.95
fisicaI_202 2-23_06.xls	9.68	43.86	0.07	46.39	53.54	53.61
fisicaI_202 2- 23_07.xlsx	0.32	25.54	74.14	0.0	25.86	100.0
fisicaI_202 2-23_08.xls	44.25	45.79	7.26	2.7	90.04	97.3
fisicaI_202 2-23_09.xls	5.23	82.29	12.46	0.03	87.51	99.97
fisicaI_202 2-23_11.xls	16.96	63.8	13.1	6.14	80.76	93.86

fisicaI_202 2-23_12.xls	69.51	16.05	7.97	6.47	85.56	93.53
fisicaI_202 2- 23_13.xlsx	33.49	64.96	0.21	1.34	98.45	98.66
fisicaI_202 2-23_14.xls	49.43	18.57	16.53	15.47	68.0	84.53
fisicaI_202 2- 23_15.xlsx	97.79	0.58	0.35	1.28	98.38	98.72
fisicaI_202 2-23_17.xls	10.49	32.68	1.02	55.81	43.17	44.19
fisicaI_202 2-23_18.xls	17.0	56.64	16.67	9.69	73.64	90.31
fisicaI_202 2-23_19.xls	54.89	8.02	24.44	12.66	62.91	87.34
fisicaI_202 2-23_20.xls	32.0	2.82	64.68	0.51	34.82	99.49
fisicaI_202 2-23_21.xls	12.39	67.33	7.55	12.74	79.71	87.26
fisicaI_202 2- 23_22.xlsx	10.28	58.99	12.07	18.66	69.27	81.34
fisicaI_202 2-23_23.xls	43.47	29.27	6.81	20.46	72.73	79.54
fisicaI_202 2-23_24.xls	0.17	6.05	81.78	11.99	6.23	88.01
fisicaI_202 2-23_25.xls	22.27	2.63	33.58	41.52	24.9	58.48
fisicaI_202 2-23_26.xls	9.44	26.66	63.89	0.01	36.1	99.99
fisicaI_202 2-23_27.xls	79.09	0.13	20.76	0.01	79.23	99.99
fisicaI_202	0.77	19.36	0.9	78.97	20.13	21.03

2-23_30.xls						
fisicaI_202 2-23_31.xls	8.89	27.72	33.49	29.9	36.62	70.1
fisicaI_202 2- 23_33.xlsx	33.98	15.25	49.09	1.67	49.24	98.33
fisicaI_202 2-23_34.xls	8.14	48.42	39.49	3.95	56.56	96.05
fisicaI_202 2-23_35.xls	4.73	24.45	70.61	0.21	29.18	99.79
fisicaI_202 2-23_36.xls	11.98	1.59	1.65	84.79	13.57	15.21
fisicaI_202 2-23_37.xls	26.1	33.4	6.03	34.47	59.5	65.53
fisicaI_202 2- 23_38.xlsx	70.78	0.01	29.13	0.08	70.79	99.92
fisicaI_202 2-23_39.xls	8.43	91.11	0.13	0.34	99.54	99.66
fisicaI_202 2-23_40.xls	19.22	1.64	0.18	78.96	20.86	21.04
fisicaI_202 2-23_41.xls	60.37	0.42	39.1	0.11	60.79	99.89
fisicaI_202 2-23_42.xls	32.91	30.42	19.22	17.45	63.33	82.55
fisicaI_202 2-23_43.xls	1.99	89.64	2.97	5.39	91.63	94.61
fisicaI_202 2-23_44.xls	26.64	56.34	4.83	12.19	82.98	87.81
fisicaI_202 2- 23_45.xlsx	1.9	36.95	3.75	57.4	38.84	42.6

fisicaI_202 2-23_46.xls	2.76	54.33	19.25	23.66	57.09	76.34
fisicaI_202 2-23_47.xls	65.15	3.18	30.65	1.02	68.33	98.98
fisicaI_202 2-23_48.xls	47.21	50.32	2.2	0.27	97.54	99.73
fisicaI_202 2-23_49.xls	2.44	9.98	81.82	5.77	12.42	94.23
fisicaI_202 2-23_50.xls	1.45	97.49	0.52	0.53	98.94	99.47
fisicaI_202 2-23_51.xls	26.28	72.69	0.21	0.81	98.98	99.19
fisicaI_202 2-23_52.xls	20.47	76.63	0.0	2.91	97.09	97.09
fisicaI_202 2- 23_53.xlsx	75.36	0.41	0.01	24.23	75.76	75.77
fisicaI_202 2- 23_54.xlsx	32.91	30.42	19.22	17.45	63.33	82.55
fisicaI_202 2-23_55.xls	0.49	3.4	25.8	70.31	3.9	29.69
fisicaI_202 2-23_56.xls	3.41	0.0	63.98	32.61	3.41	67.39
fisicaI_202 2-23_57.xls	0.47	65.82	31.68	2.03	66.29	97.97
fisicaI_202 2- 23_58.xlsx	46.9	7.99	10.55	34.56	54.88	65.44
fisicaI_202 2-23_59.xls	42.1	56.01	0.46	1.42	98.12	98.58
fisicaI_202 2-23_61.xls	53.45	43.97	1.4	1.17	97.42	98.83
fisicaI_202	20.51	58.14	21.36	0.0	78.64	100.0

2-23_62.xls						
fisicaI_202 2- 23_63.xlsx	3.99	58.14	17.65	20.22	62.13	79.78
fisicaI_202 2- 23_64.xlsx	92.49	2.94	2.42	2.15	95.44	97.85
fisicaI_202 2-23_65.xls	1.54	13.7	78.89	5.86	15.25	94.14
fisicaI_202 2-23_66.xls	72.79	17.1	9.23	0.89	89.88	99.11
fisicaI_202 2-23_67.xls	0.3	52.28	7.39	40.03	52.58	59.97
fisicaI_202 2-23_68.xls	32.13	14.69	43.59	9.58	46.82	90.42
fisicaI_202 2-23_69.xls	0.38	84.88	7.33	7.41	85.26	92.59
fisicaI_202 2-23_70.xls	62.35	17.48	20.17	0.0	79.83	100.0
fisicaI_202 2-23_71.xls	97.92	0.11	1.97	0.01	98.03	99.99
fisicaI_202 2-23_72.xls	49.23	0.2	13.76	36.81	49.44	63.19
fisicaI_202 2-23_73.xls	87.57	0.25	8.76	3.43	87.82	96.57
fisicaI_202 2- 23_74.xlsx	26.78	55.77	0.82	16.64	82.54	83.36
fisicaI_202 2-23_75.xls	27.94	8.21	3.51	60.34	36.15	39.66
fisicaI_202 2-23_76.xls	52.14	47.39	0.46	0.01	99.53	99.99

fisicaI_202 3-24_01.xls	1.9	44.09	30.05	23.96	45.99	76.04
fisicaI_202 3-24_02.xls	69.38	6.1	20.96	3.56	75.48	96.44
fisicaI_202 3- 24_03.xlsx	87.08	0.62	0.41	11.9	87.69	88.1
fisicaI_202 3-24_04.xls	10.06	68.97	19.83	1.14	79.03	98.86
fisicaI_202 3-24_05.xls	23.05	64.0	5.45	7.5	87.04	92.5
fisicaI_202 3- 24_06.xlsx	48.28	49.04	2.51	0.17	97.33	99.83
fisicaI_202 3- 24_07.xlsx	31.8	66.55	1.65	0.0	98.34	100.0
fisicaI_202 3-24_08.xls	10.2	85.12	2.06	2.62	95.31	97.38
fisicaI_202 3-24_09.xls	54.12	28.61	12.73	4.54	82.73	95.46
fisicaI_202 3-24_10.xls	0.87	63.89	5.85	29.39	64.76	70.61
fisicaI_202 3- 24_11.xlsx	3.87	93.25	0.45	2.43	97.13	97.57
fisicaI_202 3-24_12.xls	1.54	13.7	78.89	5.86	15.25	94.14
fisicaI_202 3-24_13.xls	15.75	16.75	64.56	2.95	32.49	97.05
fisicaI_202 3-24_14.xls	61.08	0.3	20.11	18.52	61.37	81.48
fisicaI_202 3-24_15.xls	10.44	26.44	56.47	6.65	36.89	93.35

fisicaI_202 3-24_16.xls	15.38	75.75	4.99	3.88	91.13	96.12
fisicaI_202 3-24_17.xls	3.69	7.06	42.06	47.2	10.75	52.8
fisicaI_202 3-24_18.xls	45.2	9.15	32.61	13.04	54.35	86.96
fisicaI_202 3-24_20.xls	0.2	93.38	6.05	0.37	93.59	99.63
fisicaI_202 3-24_21.xls	0.76	97.73	0.68	0.84	98.48	99.16
fisicaI_202 3-24_22.xls	36.09	0.06	21.25	42.61	36.15	57.39
fisicaI_202 3-24_23.xls	22.79	64.0	0.11	13.1	86.78	86.9
fisicaI_202 3-24_24.xls	6.12	69.97	0.02	23.89	76.09	76.11
fisicaI_202 3-24_25.xls	7.56	60.26	19.87	12.31	67.82	87.69
fisicaI_202 3-24_26.xls	2.48	77.36	4.6	15.57	79.84	84.43
fisicaI_202 3-24_27.xls	95.57	3.13	1.11	0.19	98.7	99.81
fisicaI_202 3-24_28.xls	14.03	84.81	0.59	0.57	98.83	99.43
fisicaI_202 3- 24_29.xlsx	89.23	3.78	6.98	0.02	93.0	99.98
fisicaI_202 3-24_30.xls	61.67	9.31	16.0	13.02	70.98	86.98
fisicaI_202 3-24_31.xls	68.2	26.69	1.2	3.92	94.89	96.08
fisicaI_202 3-24_32.xls	0.0	0.28	25.57	74.15	0.28	25.85

fisicaI_202 3-24_33.xls	27.58	38.78	32.75	0.89	66.37	99.11
fisicaI_202 3-24_34.xls	28.35	56.24	14.85	0.55	84.59	99.45
fisicaI_202 3-24_35.xls	3.95	7.36	76.24	12.45	11.31	87.55
fisicaI_202 3-24_36.xls	83.34	1.82	11.06	3.78	85.16	96.22
fisicaI_202 3-24_37.xls	36.23	50.21	9.35	4.21	86.43	95.79
fisicaI_202 3- 24_38.xlsx	10.05	70.39	12.81	6.75	80.44	93.25
fisicaI_202 3- 24_39.xlsx	59.65	10.17	30.16	0.02	69.82	99.98
fisicaI_202 3-24_40.xls	28.58	45.56	19.69	6.17	74.14	93.83
fisicaI_202 3-24_41.xls	8.92	78.66	1.27	11.15	87.58	88.85
fisicaI_202 3-24_42.xls	9.09	90.78	0.04	0.09	99.87	99.91
fisicaI_202 3-24_43.xls	6.36	67.11	25.39	1.15	73.46	98.85
fisicaI_202 3- 24_44.xlsx	10.4	89.34	0.04	0.22	99.74	99.78
fisicaI_202 3-24_45.xls	43.99	31.96	8.96	15.1	75.95	84.9
fisicaI_202 3-24_46.xls	0.07	87.06	8.39	4.48	87.13	95.52
fisicaI_202 3-24_47.xls	0.91	1.29	93.43	4.36	2.21	95.64
fisicaI_202	3.46	90.72	1.97	3.85	94.18	96.15

3-24_48.xls						
fisicaI_202 3- 24_49.xlsx	7.76	10.09	13.92	68.23	17.85	31.77
fisicaI_202 3- 24_50.xlsx	0.0	97.6	2.37	0.03	97.6	99.97
fisicaI_202 3-24_51.xls	1.99	22.27	62.18	13.56	24.26	86.44
fisicaI_202 3- 24_52.xlsx	57.05	1.92	37.17	3.85	58.98	96.15
fisicaI_202 3- 24_53.xlsx	78.97	19.01	1.65	0.38	97.97	99.62
fisicaI_202 3- 24_54.xlsx	40.29	2.51	35.78	21.42	42.8	78.58
fisicaI_202 3-24_55.xls	24.01	18.34	56.39	1.26	42.36	98.74
fisicaI_202 3-24_56.xls	9.55	1.53	1.04	87.88	11.08	12.12
fisicaI_202 3- 24_57.xlsx	35.39	9.8	16.79	38.02	45.19	61.98
fisicaI_202 3-24_58.xls	63.56	33.75	2.09	0.6	97.31	99.4
fisicaI_202 3-24_59.xls	96.33	3.1	0.57	0.0	99.43	100.0
fisicaI_202 3- 24_60.xlsx	97.04	0.74	2.21	0.01	97.78	99.99
fisicaI_202 4-25_01.xls	66.84	26.76	4.08	2.32	93.6	97.68
fisicaI_202	87.21	8.35	4.11	0.33	95.56	99.67

4-25_02.xls						
fisicaI_202 4- 25_03.xlsx	81.71	13.07	0.94	4.28	94.78	95.72
fisicaI_202 4-25_04.xls	1.08	33.86	16.96	48.11	34.93	51.89
fisicaI_202 4-25_05.xls	31.23	32.18	27.82	8.77	63.41	91.23
fisicaI_202 4-25_06.xls	4.74	82.73	11.07	1.46	87.47	98.54
fisicaI_202 4-25_07.xls	11.79	87.68	0.33	0.2	99.47	99.8
fisicaI_202 4-25_08.xls	24.25	64.49	5.2	6.05	88.75	93.95
fisicaI_202 4-25_09.xls	51.73	33.71	1.05	13.51	85.44	86.49
fisicaI_202 4-25_10.xls	11.17	13.64	7.55	67.64	24.81	32.36
fisicaI_202 4- 25_100.xls	43.99	32.19	6.64	17.18	76.18	82.82
fisicaI_202 4- 25_101.xls	6.88	3.8	88.38	0.94	10.68	99.06
fisicaI_202 4- 25_102.xlsx	84.67	12.73	1.79	0.82	97.39	99.18
fisicaI_202 4- 25_103.xls	65.93	5.55	1.68	26.84	71.48	73.16
fisicaI_202 4- 25_104.xls	8.15	42.98	24.02	24.85	51.13	75.15
fisicaI_202 4-	41.27	29.31	23.01	6.41	70.58	93.59

25_105.xls

fisicaI_202 4-	2.45	63.68	1.85	32.02	66.13	67.98
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25_11.xlsx

fisicaI_202 4-	34.93	61.47	2.55	1.06	96.4	98.94
-------------------	-------	-------	------	------	------	-------

25_12.xlsx

fisicaI_202 4-25_13.xls	1.69	0.85	97.44	0.02	2.54	99.98
----------------------------	------	------	-------	------	------	-------

fisicaI_202 4-	12.35	36.56	45.9	5.19	48.91	94.81
-------------------	-------	-------	------	------	-------	-------

25_14.xlsx

fisicaI_202 4-25_15.xls	11.45	37.69	47.35	3.51	49.14	96.49
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2.3 Global analysis of the principal components

Figure 6 shows a four-dimensional representation of the learning styles of the students as a function of the contributions to three of the CHAEA learning styles. The tendency towards the remaining learning style is implicitly shown in the size and color shades of the points (bigger and darker colors imply a larger tendency). The principal directions shown as continuous lines (PC0, red; PC1, green; PC2, purple; PC3, orange) emerge from centroid, which is given by the average mean of the data set and is shown as a black star. The projection of the points and the average mean is shown in gray. The projection confidence interval of the average mean is also shown. The origin, which is defined as (0, 0, 0, 0) in the 4D space of the learning styles is shown as a black triangle. The colored points correspond to the maximal pure learning styles (e.g., (20, 0, 0, 0) for activist (red), or (0, 20, 0, 0) for reflector (green)) are also marked.

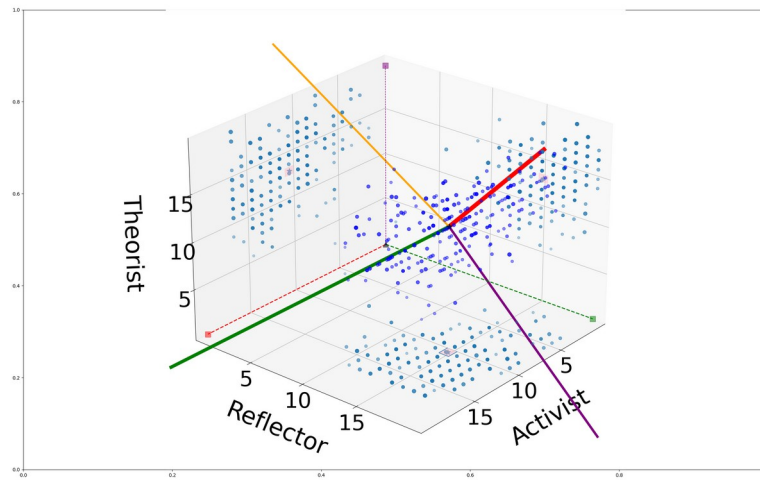


Figure 6. Four-dimensional representation of the learning styles of the students as a function of the CHAEA learning styles.

The central panel in Fig. 7 shows the projection of the learning styles of the student represented in Fig. 6 on the two main principal components (PC0 and PC1), along with their histograms. The data surround the average mean (gray star). The points (20, 0, 0, 0), (0, 20, 0, 0), (0, 0, 20, 0), and (0, 0, 0, 20), which correspond, respectively, to the maximal pure activist (red), reflector (green), theorist (purple), and pragmatist (orange) learning styles in the original basis set are also marked, and joined with a dashed line to the corresponding origin (0, 0, 0, 0) (black triangle). The point (20, 20, 20, 20) is also shown (brown diamond).

The reduced dimensional representation of the data given by Fig. 6 is usually meaningful if $\Sigma_1(\%) \geq 70.0\%$; in the case under study we have that $\Sigma_1(\%) = 93.5$. The top and left panels of Fig. 7 show the histograms of the data as a function of PC0 (red) and PC1 (green), which are fitted using Weibull distributions with the parameters listed in Table 9.

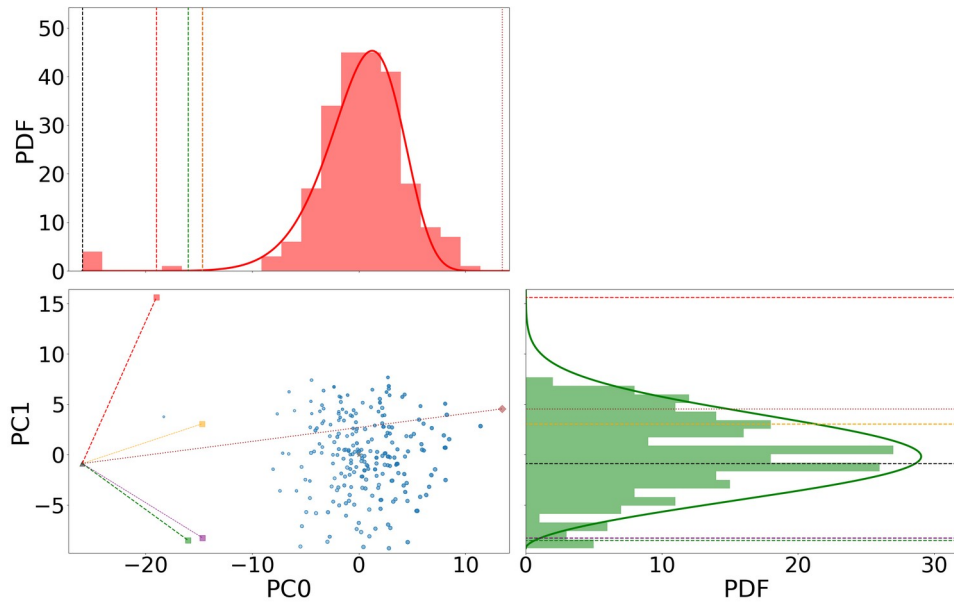


Figure 7. The projection of the learning styles of the students represented in Fig. 6 on the two main principal components (PC0 and PC1), along with their histograms as a function of PC0 (red) and PC1 (green).

Table 9. Parameters of the Weibull distributions that fit the probability distributions (histograms) of the projection of the learning styles on the principal-components basis set (see histograms in Fig. 7). The location parameter θ is set equal to the smallest projection for each component.

Principal component	α	k	θ
0	27.58	8.6	-25.94
1	10.66	2.8	-9.28
2	8.48	3.85	-7.69
3	7.6	5.02	-7.06

3. Participation ratios

Third, the participation ratios are presented in this section. The values for the CHAEA (activist, reflector, theorist, and pragmatist), and for the principal-components basis sets are listed. A statistical analysis of the distribution of the participation ratios is also included.

3.1 Individual analysis of the participation ratios

Table 10 shows the participation ratios of each individual student for CHAEA (activist, reflector, theorist, and pragmatist) and for the principal-components basis sets. For the case under study, this parameter lies between 1 and 4. The smaller the participation ratio, the better.

Table 10. Participation ratios of each individual student for CHAEA learning styles (LS) and for the principal-components (PC) basis sets.

Student	LS	PC
fisicaI_2022-23_01.xls	3.98	2.22
fisicaI_2022-23_02.xls	3.31	1.08
fisicaI_2022-23_03.xls	3.94	2.72
fisicaI_2022-23_04.xlsx	2.96	3.38
fisicaI_2022-23_05.xlsx	3.63	2.68
fisicaI_2022-23_06.xls	3.71	2.4
fisicaI_2022-23_07.xlsx	3.5	1.63
fisicaI_2022-23_08.xls	3.21	2.43
fisicaI_2022-23_09.xls	3.22	1.44
fisicaI_2022-23_11.xls	3.85	2.19
fisicaI_2022-23_12.xls	3.89	1.92
fisicaI_2022-23_13.xlsx	3.84	1.87
fisicaI_2022-23_14.xls	2.31	3.03
fisicaI_2022-23_15.xlsx	3.8	1.05
fisicaI_2022-23_17.xls	3.48	2.33
fisicaI_2022-23_18.xls	3.19	2.58
fisicaI_2022-23_19.xls	3.65	2.61
fisicaI_2022-23_20.xls	3.86	1.92

fisicaI_2022-23_21.xls	3.77	2.04
fisicaI_2022-23_22.xlsx	3.42	2.45
fisicaI_2022-23_23.xls	3.44	3.11
fisicaI_2022-23_24.xls	3.59	1.46
fisicaI_2022-23_25.xls	3.7	2.98
fisicaI_2022-23_26.xls	3.61	2.05
fisicaI_2022-23_27.xls	3.65	1.5
fisicaI_2022-23_30.xls	3.49	1.51
fisicaI_2022-23_31.xls	3.34	3.49
fisicaI_2022-23_33.xlsx	3.56	2.63
fisicaI_2022-23_34.xls	3.21	2.51
fisicaI_2022-23_35.xls	2.9	1.78
fisicaI_2022-23_36.xls	3.63	1.36
fisicaI_2022-23_37.xls	3.48	3.31
fisicaI_2022-23_38.xlsx	2.86	1.71
fisicaI_2022-23_39.xls	2.81	1.19
fisicaI_2022-23_40.xls	3.03	1.51
fisicaI_2022-23_41.xls	3.75	1.93
fisicaI_2022-23_42.xls	3.58	3.73
fisicaI_2022-23_43.xls	3.54	1.24
fisicaI_2022-23_44.xls	3.89	2.47
fisicaI_2022-23_45.xlsx	3.89	2.14
fisicaI_2022-23_46.xls	3.04	2.57
fisicaI_2022-23_47.xls	3.5	1.92
fisicaI_2022-23_48.xls	3.24	2.1

fisicaI_2022-23_49.xls	3.75	1.46
fisicaI_2022-23_50.xls	2.77	1.05
fisicaI_2022-23_51.xls	2.4	1.67
fisicaI_2022-23_52.xls	3.08	1.59
fisicaI_2022-23_53.xlsx	3.67	1.6
fisicaI_2022-23_54.xlsx	3.58	3.73
fisicaI_2022-23_55.xls	3.58	1.78
fisicaI_2022-23_56.xls	3.16	1.93
fisicaI_2022-23_57.xls	3.95	1.87
fisicaI_2022-23_58.xlsx	3.64	2.8
fisicaI_2022-23_59.xls	2.78	2.04
fisicaI_2022-23_61.xls	3.92	2.09
fisicaI_2022-23_62.xls	3.42	2.35
fisicaI_2022-23_63.xlsx	3.52	2.43
fisicaI_2022-23_64.xlsx	3.06	1.17
fisicaI_2022-23_65.xls	3.54	1.55
fisicaI_2022-23_66.xls	2.51	1.76
fisicaI_2022-23_67.xls	3.78	2.28
fisicaI_2022-23_68.xls	3.54	3.09
fisicaI_2022-23_69.xls	3.85	1.37
fisicaI_2022-23_70.xls	2.94	2.17
fisicaI_2022-23_71.xls	1.0	1.04
fisicaI_2022-23_72.xls	3.71	2.52
fisicaI_2022-23_73.xls	3.76	1.29
fisicaI_2022-23_74.xlsx	2.73	2.44
fisicaI_2022-23_75.xls	3.79	2.22

fisicaI_2022-23_76.xls	3.26	2.01
fisicaI_2023-24_01.xls	3.63	2.92
fisicaI_2023-24_02.xls	3.51	1.89
fisicaI_2023-24_03.xlsx	3.85	1.29
fisicaI_2023-24_04.xls	3.71	1.9
fisicaI_2023-24_05.xls	2.94	2.12
fisicaI_2023-24_06.xlsx	3.23	2.11
fisicaI_2023-24_07.xlsx	3.3	1.84
fisicaI_2023-24_08.xls	3.52	1.36
fisicaI_2023-24_09.xls	3.91	2.54
fisicaI_2023-24_10.xls	3.41	2.01
fisicaI_2023-24_11.xlsx	3.18	1.15
fisicaI_2023-24_12.xls	3.54	1.55
fisicaI_2023-24_13.xls	2.86	2.13
fisicaI_2023-24_14.xls	3.79	2.23
fisicaI_2023-24_15.xls	3.04	2.47
fisicaI_2023-24_16.xls	3.14	1.66
fisicaI_2023-24_17.xls	3.48	2.46
fisicaI_2023-24_18.xls	3.47	2.98
fisicaI_2023-24_20.xls	2.8	1.14
fisicaI_2023-24_21.xls	3.09	1.05
fisicaI_2023-24_22.xls	3.22	2.8
fisicaI_2023-24_23.xls	3.52	2.09
fisicaI_2023-24_24.xls	3.51	1.82
fisicaI_2023-24_25.xls	3.77	2.36
fisicaI_2023-24_26.xls	2.98	1.6

fisicaI_2023-24_27.xls	3.92	1.09
fisicaI_2023-24_28.xls	2.92	1.35
fisicaI_2023-24_29.xlsx	2.51	1.25
fisicaI_2023-24_30.xls	3.84	2.32
fisicaI_2023-24_31.xls	3.83	1.86
fisicaI_2023-24_32.xls	3.25	1.63
fisicaI_2023-24_33.xls	3.8	3.0
fisicaI_2023-24_34.xls	3.76	2.39
fisicaI_2023-24_35.xls	3.18	1.66
fisicaI_2023-24_36.xls	3.83	1.41
fisicaI_2023-24_37.xls	3.41	2.54
fisicaI_2023-24_38.xlsx	3.23	1.9
fisicaI_2023-24_39.xlsx	3.71	2.19
fisicaI_2023-24_40.xls	3.42	3.01
fisicaI_2023-24_41.xls	2.76	1.56
fisicaI_2023-24_42.xls	3.7	1.2
fisicaI_2023-24_43.xls	3.07	1.93
fisicaI_2023-24_44.xlsx	2.59	1.24
fisicaI_2023-24_45.xls	3.37	3.06
fisicaI_2023-24_46.xls	2.58	1.3
fisicaI_2023-24_47.xls	3.73	1.14
fisicaI_2023-24_48.xls	3.69	1.21
fisicaI_2023-24_49.xlsx	3.59	2.0
fisicaI_2023-24_50.xlsx	2.99	1.05
fisicaI_2023-24_51.xls	3.45	2.2
fisicaI_2023-24_52.xlsx	3.81	2.15

fisicaI_2023-24_53.xlsx	3.6	1.52
fisicaI_2023-24_54.xlsx	3.82	2.97
fisicaI_2023-24_55.xls	3.71	2.44
fisicaI_2023-24_56.xls	3.7	1.28
fisicaI_2023-24_57.xlsx	3.73	3.25
fisicaI_2023-24_58.xls	3.4	1.93
fisicaI_2023-24_59.xls	3.88	1.08
fisicaI_2023-24_60.xlsx	3.86	1.06
fisicaI_2024-25_01.xls	3.42	1.92
fisicaI_2024-25_02.xls	3.58	1.3
fisicaI_2024-25_03.xlsx	3.96	1.46
fisicaI_2024-25_04.xls	3.82	2.67
fisicaI_2024-25_05.xls	3.42	3.49
fisicaI_2024-25_06.xls	3.81	1.43
fisicaI_2024-25_07.xls	3.09	1.28
fisicaI_2024-25_08.xls	3.85	2.08
fisicaI_2024-25_09.xls	3.98	2.5
fisicaI_2024-25_10.xls	3.64	2.02
fisicaI_2024-25_100.xls	3.47	3.02
fisicaI_2024-25_101.xls	3.41	1.27
fisicaI_2024-25_102.xlsx	3.93	1.36
fisicaI_2024-25_103.xls	3.59	1.96
fisicaI_2024-25_104.xls	3.29	3.22
fisicaI_2024-25_105.xls	3.36	3.19
fisicaI_2024-25_11.xlsx	3.82	1.96
fisicaI_2024-25_12.xlsx	2.48	2.0

fisicaI_2024-25_13.xls	3.49	1.05
fisicaI_2024-25_14.xlsx	2.82	2.76
fisicaI_2024-25_15.xls	3.42	2.63

3.2 Global analysis of the participation ratios

Figure 8 shows the probability distribution functions of the participation ratios for the learning-styles basis set (red) and for the principal components (blue). The vertical dashed lines mark the average means along with their corresponding confidence intervals (shaded areas), whose values can be found in Table 11. The corresponding cumulative distributions used in the fitting are shown in Fig. 9.

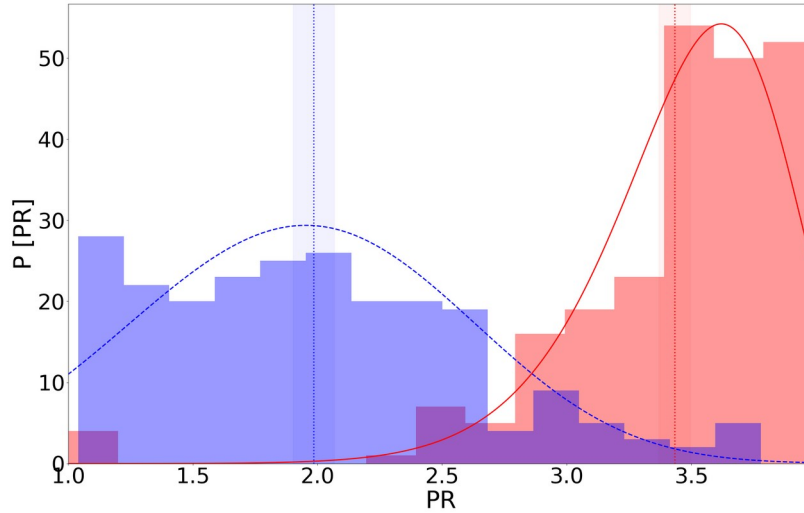


Figure 8. Probability distribution functions of the participation ratios associated with the learning-styles basis set (red) and with the principal components (blue). The vertical dashed lines mark the average means along with their corresponding confidence intervals (shaded areas), whose values can be found in Table 11. The continuous lines show fitting functions given by Weibull distributions with parameters shown in Table 11.

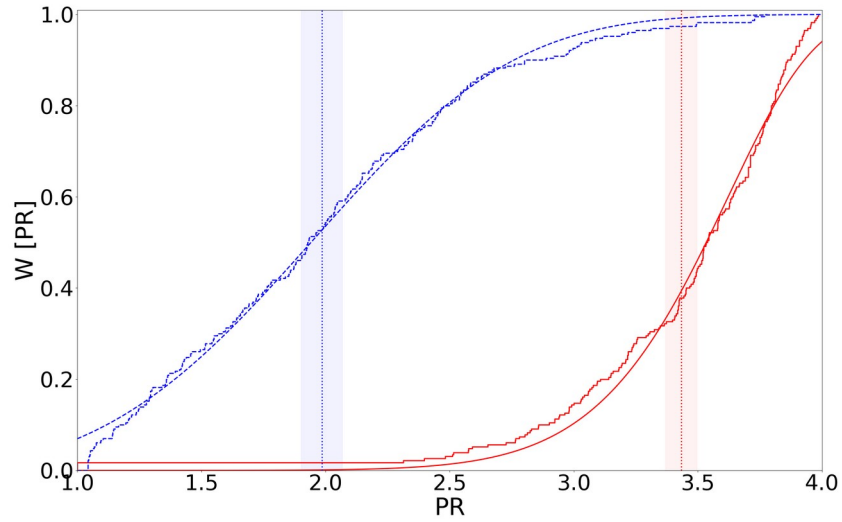


Figure 9. Cumulative distributions for the results of Fig. 8.

Table 11. Average mean and corresponding uncertainty (in parenthesis) of the participation ratios for the learning-styles basis set and for the principal components. α and k are, respectively, the shape and scale parameters of the Weibull distributions the fit the probability distributions (histograms) of Fig. 9. The location parameter is set equal to $\theta = 0$.

Basis set	Average mean (Uncertainty)	α	k
Learning styles	3.43(0.06)	3.65	11.33
Principal components	1.99(0.08)	2.16	3.42

4. Students characteristics and recommendations

This section summarizes some of the characteristics associated with the activist, theorist, reflector, and pragmatist learning styles, and provides some recommendation to increase the tendencies towards them, as detailed by Alonso, Gallego & Honey (2007), and Cancino (2014).

Alonso, C. M., Gallego, D. J., & Honey, P. (2007). *Los estilos de aprendizaje. Procedimiento de diagnóstico y mejora* (7th ed.). Ediciones Mensajero, S. A. U.

Cancino, O. (2024, March). *Blog de la prof de Français: espacio para intercambiar con los estudiantes*. add_hyperlink(p, "<https://oticar.wordpress.com/estilos-de-aprendizaje/>", "<https://oticar.wordpress.com/estilos-de-aprendizaje/>")

4.1 Affinity and dominant learning style

According to the affinity, the dominant learning style is the theorist. The value of its affinity is equal to 87.9 %.

The main characteristics of the theorist learning style are:

- Methodical
- Logical
- Objective
- Critical
- Structured

Furthermore, other characteristics usually associated with the theorist learning style are being: disciplined, planned, systematic, organized, synthetic, reasoner, thinker, connector, perfectionist, generalizer, hypothesis-seeker, theory-seeker, model-seeker, question-asker, assumption-explorer, concept-seeker, clear-purpose seeker, rationality-seeker, "why"-asker, value-system and criteria-seeker, procedure-inventor, explorer.

According to the principal components analysis, the differences of the students' learning styles mostly correspond to the theorist learning style, as 31.8 % of the eigenvector associated with the main principal component (PC0) is projected onto it.

Activist learning style

The activist learning style has an average and an uncertainty (in parenthesis) of 11.0(0.4) points.

People with scores higher than 13 on the test find more comfortable in learning situations where they can:

- Attempt new things, new experiences, new opportunities.
- Compete in a team.
- Generate ideas without formal or structural limitations.
- Solve problems.
- Change and vary things.
- Address multiple tasks.
- Engage in dramatization.

- Play roles.
- Experience situations of interest or crisis.
- Capture attention.

Similarly, they'll feel uncomfortable in situations that demand:

- Presenting topics with a heavy theoretical load: Explaining causes, backgrounds...
- Assimilating, analyzing, and interpreting many unclear data.
- Paying attention to details.
- Working alone, reading, writing, or thinking alone.
- Evaluating in advance what is going to be learned.
- Assessing what has already been achieved or learned.
- Repeating the same activity.

If someone has scored below 9, it's advisable to develop certain aspects that characterize this learning style further. Suggestions for improvement include:

- Trying something new, something never done before, at least once a week... Bringing something attention-grabbing to the study or workplace. Reading a newspaper with opposing opinions. Rearranging furniture at home or work.
- Practicing initiating conversations with strangers. In large gatherings, parties, conferences, forcing oneself to initiate and sustain conversations with everyone present, if possible. During leisure time, trying to engage in dialogue with strangers or convince them of our ideas.
- Deliberately fragmenting the day by cutting and changing activities every half an hour...

Blocks that inhibit the development of the activist style:

- Fear of failure or making mistakes.
- Fear of ridicule.
- Anxiety towards new or unfamiliar things.
- Strong desire to thoroughly think things through in advance.
- Self-doubt, lack of self-confidence.
- Taking life very seriously, very conscientiously.

Reflector learning style

The reflector learning style has an average and an uncertainty (in parenthesis) of 15.2(0.4) points.

People with scores higher than 18 on the test learn better in learning situations where they can:

- Observe.
- Reflect on activities.
- Exchange opinions with others by prior agreement.

- Reach decisions at their own pace.
- Work without pressures or mandatory deadlines.
- Review what's been learned, what has happened.
- Investigate thoroughly.
- Gather information.
- Probe to get to the heart of the matter.
- Think before acting.

Similarly, they'll feel uncomfortable in situations that demand:

- Taking the forefront.
- Acting as a leader.
- Chairing meetings or debates.
- Performing in front of observing individuals.
- Playing a role.
- Participating in situations that require action without planning.
- Doing something without prior notice. Presenting an idea spontaneously.

If someone has scored below 14, they might consider the following suggestions to enhance the reflector style:

- Practice observation. Study people's behavior. Note who speaks the most, who interrupts, how often the teacher summarizes... Study non-verbal behavior, when people look at the clock, cross their arms, bite their pencil...
- Keep a personal diary. Reflect on the day's events and see if any conclusions can be drawn from them.
- Practice review after a meeting or event. Review the sequence of events, what went well, what could be improved. Record a dialogue or conversation on a tape recorder and play it back at least twice. Make a list of lessons learned in this way.

Blocks that hinder the development of the reflector style:

- Not having enough time to plan and think.
- Preferring or enjoying quickly changing from one activity to another.
- Being impatient to start action.
- Having resistance to listening carefully and analytically.
- Resistance to presenting things in writing.

Theorist learning style

The theorist learning style has an average and an uncertainty (in parenthesis) of 13.2(0.4) points.

People with scores higher than 14 on the test prefer learning opportunities where they can:

- Feel in structured situations with a clear purpose.
- Fit all data into a system, model, concept, or theory.

- Have time to methodically explore associations and relationships between ideas, events, and situations.
- Have the chance to question.
- Participate in a question-and-answer session.
- Test methods and logic that form the basis of something.

At the same time, they'll avoid learning situations where they have to:

- Be forced to do something without a clear context or purpose.
- Participate in situations where emotions and feelings predominate.
- Engage in unstructured activities with uncertain or ambiguous purposes.
- Take part in open-ended problems.
- Act or decide without a foundation of principles, concepts, policies, or structure.

If someone has scored below 10, they might want to consider the following suggestions to enhance the theorist style:

- Read something dense that provokes thinking for 30 minutes each day (Logic, Linguistics, Theories...). Then try to summarize what they have read using their own words.
- Practice detecting inconsistencies or weaknesses in other people's arguments, in reports, in newspaper articles... Take two newspapers with different ideologies and regularly perform a comparative analysis of the differences in their perspectives.
- Take a complex situation and analyze it to pinpoint why it developed that way, what could have been done differently, and at what point. Historical or everyday life situations. Analyze how they've used their own time...

Blocks that hinder the development of the theorist style:

- Being swayed by first impressions.
- Preferring intuition and subjectivity.
- Disliking structured and organized approaches.
- Preferring spontaneity and risk-taking.

Pragmatist learning style

The pragmatist learning style has an average and an uncertainty (in parenthesis) of 12.7(0.4) points.

People with scores higher than 16 on the test find more comfortable in situations that allow them to:

- Learn techniques for doing things with evident practical advantages.
- Be exposed to a model they can emulate.
- Acquire immediately applicable techniques in your work.
- Have the immediate possibility to apply what they've learned, to experiment.
- Develop action plans with a clear result.
- Give directions, suggest shortcuts.

Likewise, they'll find difficulty in situations that demand:

- Realizing that learning isn't related to an immediate need that they recognize or cannot see.
- Perceiving that the learning doesn't have immediate importance or practical benefit.
- Learning something that is distant from reality.
- Learning theories and general principles.
- Working without clear instructions on how to do it.

If someone has scored below 10, the following suggestions will be useful for enhancing the pragmatist style:

- Gather techniques, practical ways of doing things. They can cover anything that might be useful: analytical techniques, interpersonal skills, assertiveness, time-saving presentation techniques, statistics, memory improvement techniques...
- Seek help from individuals with demonstrated experience. They can observe our technique and advise on how to improve it.
- Focus on developing action plans in meetings and discussions of all kinds. These action plans should be specific and have deadlines. Make it a rule never to leave a meeting, debate, or class without a list of actions for yourself.

Blocks that hinder the development of the pragmatist style:

- Interest in the perfect solution rather than practicality.
- Considering useful techniques as exaggerated simplifications.
- Always leaving topics open and not committing to specific actions.
- Believing that others' ideas won't work when applied to their situation.
- Enjoying marginal topics or getting lost in them.

If you have any comments or suggestions, feel free to reach out by sending an email with your feedback to fabio.revuelta@upm.es.